

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-I]

TEST # 12

DATE : 19-12-2019

REVISION TEST

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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REVISION TEST # 12

Topics

DATE : 19 - 12 - 2019

PRE-MEDICAL ENTHUSIAST COURSE : PHASE - I – PATTERN : [NEET]

PHYSICS : K.T.G., Heat Transfer, Thermodynamics

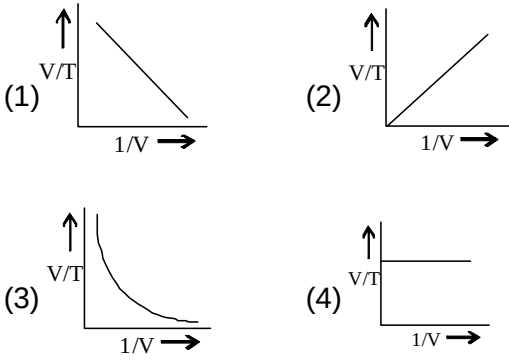
CHEMISTRY : Aldehyde and Ketone, Carboxylic Acid and Amines

BOTANY : Remaining Plant Morphology, Biotech-I

SPACE FOR ROUGH WORK

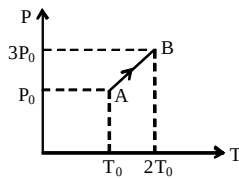
HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

- The speeds of three molecules of a gas are $3v$, $4v$ and $5v$ respectively. Their rms speed will be-
 (1) $\sqrt{\frac{50}{3}}v$ (2) $\sqrt{\frac{3}{50}}v$
 (3) $\frac{\sqrt{50}}{3}v$ (4) $4v$
- The root mean square velocity of the molecules in a sample of helium is $5/7^{\text{th}}$ that of the molecules in a sample of hydrogen. If the temperature of the hydrogen sample is 0°C , that of helium sample is about:
 (1) 0°C (2) 0 K
 (3) 273°C (4) 100°C
- If C_p and C_v are molar specific heats at constant pressure and constant volume. For hydrogen gas $C_p - C_v = a$ and for oxygen gas $C_p - C_v = b$ so the relation between a and b is -
 (1) $a = 16b$ (2) $16a = b$
 (3) $a = 4b$ (4) $a = b$
- An ideal gas is found to obey an additional law $P^2V = \text{constant}$. The gas is initially at temperature T and volume V . When it expands to a volume $2V$, the temperature becomes -
 (1) T (2) $\sqrt{2}T$
 (3) $2T$ (4) $2\sqrt{2}T$
- The correct curve between V/T and $1/V$ for a gas at constant pressure is -

- If pressure of a gas contained in a closed vessel is increased by 0.4% . When heated by 1°C the initial temperature must be
 (1) 250 K (2) 250°C
 (3) 2500 K (4) 25°C
- The kinetic energy of gas molecules at 300 K is 75 joule . This energy at 500 K will be
 (1) 125 Joule (2) 208 Joule
 (3) 270 Joule (4) 375 Joule
- Degree of freedom of hydrogen and ozone gases will be respectively -
 (1) 3 and 5 (2) 5 and 6
 (3) 6 and 5 (4) 5 and 3
- For a gas if ratio of specific heat of constant pressure and volume is γ then value of degrees of freedom is-
 (1) $\frac{3\gamma-1}{2\gamma-1}$ (2) $\frac{2}{\gamma-1}$
 (3) $\frac{9}{2}(\gamma-1)$ (4) $\frac{25}{2}(\gamma-1)$

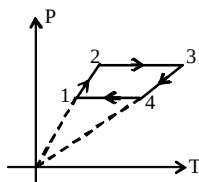
SPACE FOR ROUGH WORK

10. Pressure versus temperature graph of an ideal gas is as shown in figure. Density of the gas at point A is ρ_0 . Density at B will be –



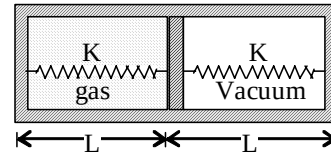
- (1) $\frac{3}{4} \rho_0$ (2) $\frac{3}{2} \rho_0$ (3) $\frac{4}{3} \rho_0$ (4) $2\rho_0$

11. P-T graph of ideal monoatomic gas is given as shown in figure. The corresponding P-V diagram is



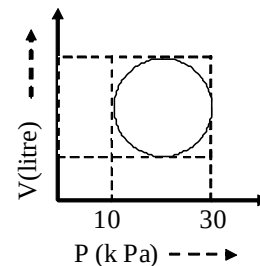
- (1) (2)
- (3) (4)

12. Area of piston is 1 m^2 . When heat is supplied to the gas it expands and displaces piston by $\frac{L}{2}$ where $L = 1\text{m}$. Natural length of springs is $L = 1\text{m}$. Spring constant $K = 100 \text{ N/m}$. The pressure of gas in final situation is –



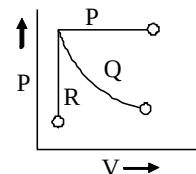
- (1) 50 N/m^2
(2) 100 N/m^2
(3) 200 N/m^2
(4) 400 N/m^2

13. Heat energy absorbed by a system in going through a cyclic process shown in figure is –



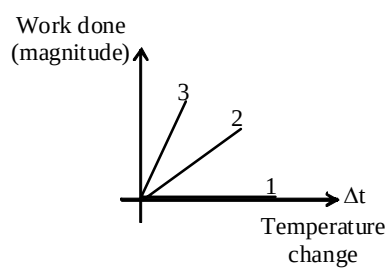
- (1) $10^7 \pi \text{ J}$ (2) $10^4 \pi \text{ J}$
(3) $10^2 \pi \text{ J}$ (4) $10^{-3} \pi \text{ J}$

14. Three curves are shown in the P-V diagram. P, Q and R represent the processes respectively



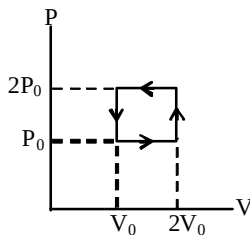
- (1) isothermal, adiabatic, isometric
(2) isobaric, isothermal, isometric
(3) isometric, isobaric, adiabatic
(4) isometric, isobaric, isothermal

SPACE FOR ROUGH WORK

15. Amount of work done in changing the state of a system from A to B is -15J . If the internal energy at A and change in internal energy 60J and $+15\text{J}$ respectively, then -
 (1) $\Delta Q = 30\text{ J}$, $U_f = 45\text{ J}$
 (2) $\Delta Q = 0$, $U_f = 75\text{ J}$
 (3) $\Delta Q = 30\text{ J}$, $U_f = 75\text{ J}$
 (4) $\Delta Q = 0$, $U_f = 45\text{ J}$
16. A polyatomic gas ($\gamma = \frac{4}{3}$) is compressed to $\frac{1}{8}$ th its volume adiabatically. If its initial pressure is P_0 , its new pressure will be -
 (1) $8 P_0$ (2) $16 P_0$
 (3) $6 P_0$ (4) $2 P_0$
17. The pressure in the tyre of a car is four times the atmospheric pressure at 300 K . If this tyre suddenly bursts, its new temperature will be ($\gamma = 1.4$)
 (1) $300(4)^{1.4/0.4}$
 (2) $300\left(\frac{1}{4}\right)^{-0.4/1.4}$
 (3) $300(2)^{-0.4/1.4}$
 (4) $300(4)^{-0.4/1.4}$
18. Work done by 0.1 mole of a gas at 27°C to double its volume at constant pressure is ($R = 2\text{ cal mol}^{-1}^\circ\text{C}^{-1}$)
 (1) 54 cal
 (2) 600 cal
 (3) 60 cal
 (4) 546 cal
19. An ideal gas at a pressures of 1 atmosphere and temperature of 27°C is compressed adiabatically until its pressure becomes 8 times the initial pressure, then the final temperature is ($\gamma = 3/2$)
 (1) 627°C
 (2) 527°C
 (3) 427°C
 (4) 327°C
20. A gas for which $\gamma = 1.5$ is suddenly compressed to $\frac{1}{4}$ th of the initial volume. Then the ratio of the final to the initial pressure is
 (1) $1 : 16$ (2) $1 : 8$
 (3) $1 : 4$ (4) $8 : 1$
21. For an ideal gas graph is shown for three processes. Processes 1, 2 and 3 are respectively—

 (1) Isochoric, isobaric, adiabatic
 (2) Isochoric, adiabatic, isobaric
 (3) Isobaric, adiabatic, isochoric
 (4) Adiabatic, isobaric, isochoric

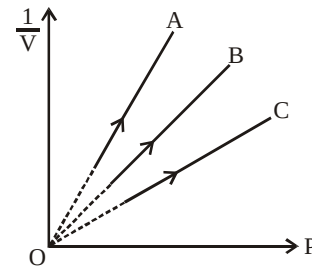
SPACE FOR ROUGH WORK

22. In Figure, an ideal gas is carried around the cyclic process. How much work is done in one cycle if $P_0 = 8 \text{ atm}$ and $V_0 = 7.00$ liters.



- (1) 5656 J (2) -5656 J
(3) 10,600 J (4) 11,300 J
23. When heat energy of 1500 Joules, is supplied to a gas at constant pressure $2.1 \times 10^5 \text{ N/m}^2$, there was an increase in its volume equal to $2.5 \times 10^{-3} \text{ m}^3$. The increase in internal energy of the gas in Joules is
(1) 450 (2) 525
(3) 975 (4) 2025
24. If heat given to a system is 6 kcal and work done is 6 kJ. Then change in internal energy is
(1) 19.1 kJ (2) 12.5 kJ
(3) 25 kJ (4) Zero
25. In an adiabatic process, the state of a gas is changed from P_1, V_1, T_1 , to P_2, V_2, T_2 . Which of the following relation is correct
(1) $T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$
(2) $P_1 V_1^{\gamma-1} = P_2 V_2^{\gamma-1}$
(3) $T_1 P_1^{\gamma} = T_2 P_2^{\gamma}$
(4) $T_1 V_1^{\gamma} = T_2 V_2^{\gamma}$

26. Figure shows the isotherms of fixed mass of an ideal gas at three temperatures T_A , T_B and T_C then.



- (1) $T_A > T_B > T_C$ (2) $T_A < T_B < T_C$
(3) $T_B < T_A < T_C$ (4) $T_A = T_B = T_C$
27. The pressure (P) and absolute temperature (T) of an ideal gas are related to each other as $P \propto \frac{1}{T^2}$. If temperature of gas is 300 K and its volume changes from V to 8 V, then find the final temperature of gas.
(1) 600 K (2) 300 K
(3) 373 K (4) 273 K
28. Four molecules of a gas have speeds 1, 2, 3 and 4 km/s. The value of the r.m.s. speed of the gas molecule is :-
(1) $\frac{1}{2}\sqrt{15}$ km/s (2) $\frac{1}{2}\sqrt{10}$ km/s
(3) 2.5 km/s (4) $\sqrt{\frac{15}{2}}$ km/s
29. The ratio of number of collisions per second at the walls of containers by He and O_2 gas molecules kept at same volume and temperature is (assume normal incidence on walls)
(1) 2 : 1 (2) 1 : 2
(3) $2\sqrt{2} : 1$ (4) $1 : 2\sqrt{2}$

SPACE FOR ROUGH WORK

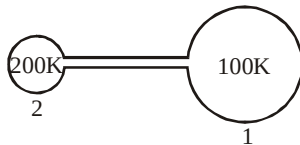
30. If hydrogen gas is heated to a very high temperature, then the fraction of energy possessed by gas molecules correspond to rotational motion :-

(1) $\frac{3}{5}$ (2) $\frac{2}{7}$ (3) $\frac{3}{7}$ (4) $\frac{2}{5}$

31. The rms speed of helium gas at 27°C and 1 atm pressure is 900 ms⁻¹. Then the rms speed of helium molecules at temperature 27°C and 2 atm pressure is

(1) 450 m/s (2) 900 m/s
(3) 1800 m/s (4) 750 m/s

32. Figure shows two flasks connected to each other. The volume of the flask 1 is twice that of flask 2. The system is filled with an ideal gas at temperature 100 K and 200 K respectively in the flasks. If the mass of the gas in 1 be m, then what is the mass of the gas in flask 2 in equilibrium?

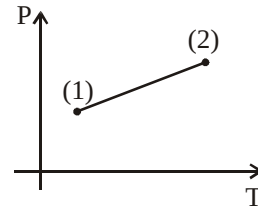


(1) m (2) $\frac{m}{2}$ (3) $\frac{m}{4}$ (4) $\frac{m}{8}$

33. A vessel contains a mixture of nitrogen of mass 7 g and carbon dioxide of mass 11 g at temperature 290 K and pressure 1 atm. The density of the mixture is :-

(1) 1.1 g/L (2) 1.2 g/L
(3) 1.51 g/L (4) 1.6 g/L

34. Figure shows a process for a given amount of an ideal gas. If volume in state 1 is V_1 & in state 2 is V_2 , then

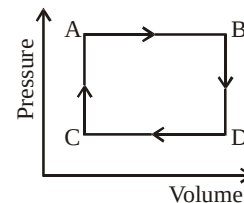


(1) $V_1 > V_2$ (2) $V_1 < V_2$
(3) $V_1 = V_2$ (4) None

35. At 20°C temperature, an argon gas at atmospheric pressure is confined in a vessel with a volume of 1m³. The effective hard-sphere diameter of argon atom is 3.10 × 10⁻¹⁰m. Its mean free path is :

(1) 100 nm (2) 90 nm
(3) 93.6 nm (4) 95 nm

36. The pressure and volume of gas are changed as shown in the P-V diagram in the figure ahead. The temperature of the gas :



(1) Increases as it goes from A to B
(2) Increases as it goes from B to C
(3) Remains constant during these changes
(4) Decreases as it goes from D to A

37. The efficiency of a Carnot's engine at a particular source and sink temperature is

$\frac{1}{2}$. When the sink temperature is reduced by 100°C, the engine efficiency, becomes

$\frac{2}{3}$. Find the source temperature.

(1) 600 K (2) 300 K
(3) 373 K (4) None

SPACE FOR ROUGH WORK

38. An ideal refrigerator runs between -23°C and 27°C . Find the heat rejected to atmosphere for every joule of work input.
(1) 6 J (2) 5 J (3) 4 J (4) 3 J

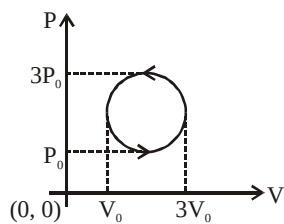
39. Select the incorrect statement about the specific heat of a gaseous system :
(1) Specific heat at no heat exchange condition, $C_A = 0$
(2) Specific heat at constant temperature, $C_T = \infty$
(3) Specific heat at constant pressure,

$$C_P = \frac{\gamma R}{\gamma - 1}$$

- (4) Specific heat at constant volume,

$$C_V = \frac{R}{\gamma}$$

40. Work done in the cyclic process shown in figure is :-



- (1) $4P_0V_0$ (2) $-4P_0V_0$
(3) $-\frac{22}{7}P_0V_0$ (4) $-13P_0V_0$

41. The specific heat of a gas in a polytropic process ($PV^x = \text{constant}$) is given by :

- (1) $C_V + \frac{R}{1+x}$ (2) $C_P + \frac{R}{1+x}$
(3) $C_V + \frac{R}{1-x}$ (4) $C_P + \frac{R}{1-x}$

42. For a certain process pressure of diatomic gas varies according to the relation " $P = aV^2$ ", where a is constant. What is the molar heat capacity of the gas for this process ?

- (1) $\frac{17R}{6}$ (2) $\frac{6R}{17}$ (3) $\frac{13R}{6}$ (4) $\frac{16R}{7}$

43. The temperature inside and outside a refrigerator are 273 K and 300 K respectively. Assuming that the refrigerator cycle is reversible. For every joule of work done heat delivered to the surrounding will be nearly :-

- (1) 11 J (2) 22 J (3) 33 J (4) 50 J

44. One mole of an ideal gas undergoes a process whose molar heat capacity is $4R$ and in which work done by gas for small change in temperature is given by the

relation $dW = 2RdT$, then the ratio $\frac{C_P}{C_V}$ is

- (1) $\frac{7}{5}$ (2) $\frac{5}{3}$ (3) $\frac{3}{2}$ (4) 2

45. A gas expands such that its initial and final temperatures are equal. Also, the process followed by the gas traces a straight line on the P - V diagram. Then
(i) the temperature of the gas remains constant throughout.

(ii) the temperature of the gas first increases and then decreases.

(iii) the temperature of the gas first decreases and then increases.

(iv) the straight line has a negative slope.

- (1) i & ii (2) i & iii
(3) ii & iv (4) ii & iii

SPACE FOR ROUGH WORK

46. A mixed salt of calcium acetate formate on dry distillation gives

- (1) ethanal
- (2) methanal
- (3) propanone
- (4) All the three above.

47. Acetic acid when heated (300°C) with MnO gives

- (1) formaldehyde
- (2) acetaldehyde
- (3) acetone
- (4) butaone

48. 'Ethylidene chloride on treatment with aq. KOH gives

- (1) CH₃CHO
- (2) (CH₂OH)₂
- (3) HCHO
- (4) (CHO)₂

49. (X) is the product of cross aldol condensation between benzaldehyde (C₆H₅CHO) and acetone. What is its structure ?

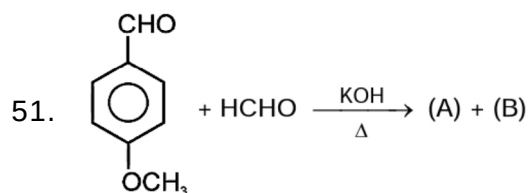
- (1) $\text{C}_6\text{H}_5\text{-CH=CH-C(=O)-CH}_3$
- (2) $\text{C}_6\text{H}_5\text{-CH=C(CH}_3)_2$
- (3) $\text{C}_6\text{H}_5\text{-CO-CH}_2\text{-C(CH}_3)_2$
- (4) None of these

50. In the reaction,

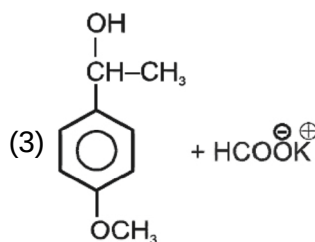
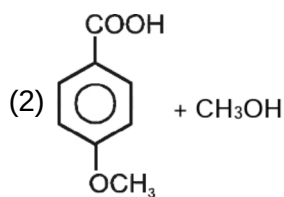
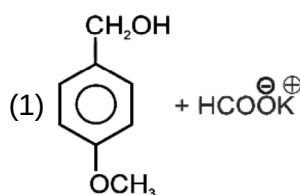


the products (A) and (B) are respectively

- (1) (CH₃)₃CCH₂OH and HCOO⁻ Na⁺.
- (2) (CH₃)₃CCOONa and CH₃OH.
- (3) (CH₃)₃CCH₂OH and CH₃OH.
- (4) (CH₃)₃COONa and HCOO⁻ Na⁺.



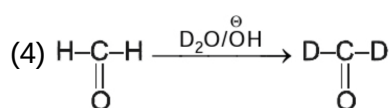
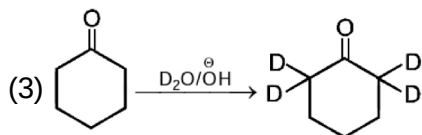
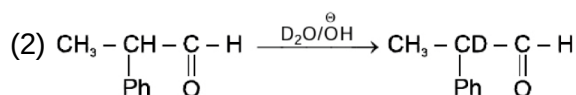
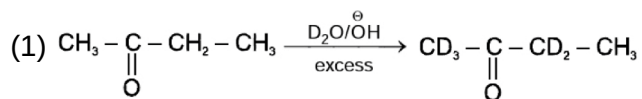
The product (A) and (B) are respectively :



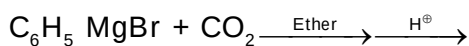
(4) Both (1) and (2)

SPACE FOR ROUGH WORK

52. In which of the following reaction deuterium exchange is not observed ?



53. In the following reaction final product is :



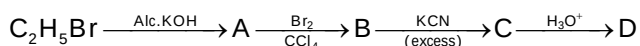
Find product

- (1) Benzoic acid (2) Benzaldehyde
(3) Benzamide (4) Benzene

54. Which of the following does not give benzoic acid salt on oxidation with hot alkaline KMnO_4 .

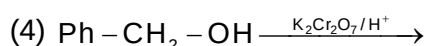
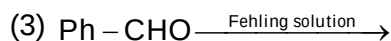
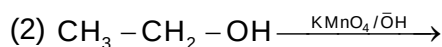
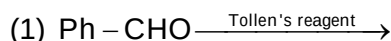
- (1) $\text{Ph}-\text{CH}_3$
(2) $\text{Ph}-\text{CH}=\text{CH}-\text{CH}_3$
(3) $\text{Ph}-\text{C}\equiv\text{C}-\text{CH}_3$
(4) $\text{Ph}-\text{C}(\text{CH}_3)_3$

55. The acid D obtained through the following sequence of reactions is



- (1) Succinic acid
(2) Malonic acid
(3) Maleic acid
(4) Oxalic acid

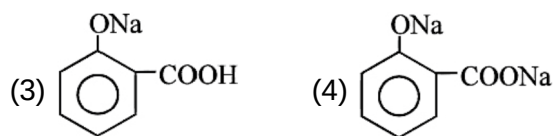
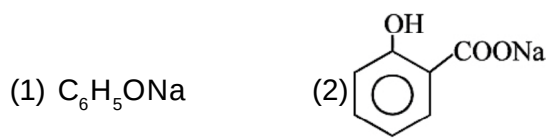
56. In which of the following reaction the final product is neither an acid nor an acid salt.



57. Formic acid can be distinguish from acetic acid because formic acid :

- (1) release H_2 with sodium
(2) gives ester with alcohol
(3) reduces ammonical AgNO_3
(4) turns red litmus to blue

58. Sodium bicarbonate reacts with salicylic acid to form:



59. Which of the following will not undergo Hell-Volhard Zelinsky (HVZ) reaction ?

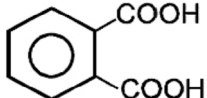
- (1) HCOOH
(2) CH_3COOH
(3) $\text{CH}_3\text{CH}_2\text{COOH}$
(4) $\text{CH}_3\text{CHBrCOOH}$

SPACE FOR ROUGH WORK

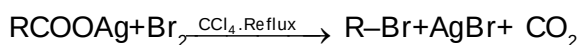
60. What product is formed when acetic acid heated with P_2O_5 .

- (1) Acetyl chloride
- (2) Acetate ester
- (3) Acetic anhydride
- (4) Acetaldehyde

61. Which of the following will not yield a cyclic compound on heating :

- (1) $\text{CH}_2 \begin{matrix} \text{COOH} \\ \text{COOH} \end{matrix}$
- (2) $\begin{matrix} \text{CH}_2 - \text{COOH} \\ | \\ \text{CH}_2 - \text{COOH} \end{matrix}$
- (3) $\text{CH}_2 \begin{matrix} \text{CH}_2\text{COOH} \\ \text{CH}_2\text{COOH} \end{matrix}$
- (4) 

62. The reaction :



is called

- (1) Wurtz reaction
- (2) Hunsdiecker bromo decarboxylation reaction
- (3) Friedel-Crafts reaction
- (4) Kolbe's reaction

63. $\text{RCOOH} \longrightarrow \text{RCH}_2\text{COOH}$.

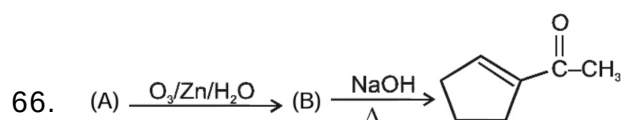
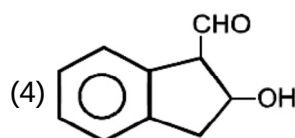
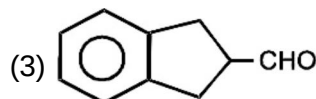
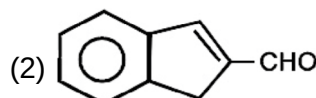
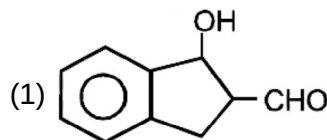
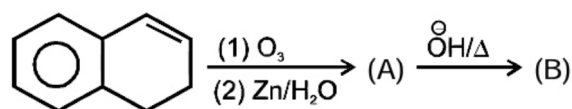
This conversion is known as reaction :

- (1) Arndt-Eistert reaction
- (2) Hunsdiecker reaction
- (3) HVZ reaction
- (4) Cannizzaro reaction

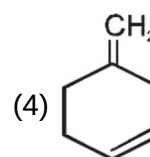
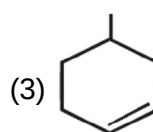
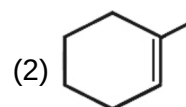
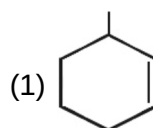
64. Acetic anhydride is prepared in the laboratory by heating sodium acetate with

- (1) ethyl chloride
- (2) acetyl chloride
- (3) cone. H_2SO_4
- (4) zinc dust

65. In the given reaction sequence B is



The reactant (A) will be :-

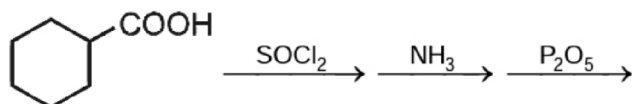


SPACE FOR ROUGH WORK

67. Which of the following does not give benzoic acid on hydrolysis ?

- (1) Phenyl cyanide
- (2) Benzoyl chloride
- (3) Benzyl chloride
- (4) Methyl benzoate

68. Identify the final product in the following sequence of reaction.



- (1)
- (2)
- (3)
- (4)

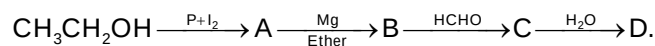
69. Which one of the following undergoes reaction with 50% sodium hydroxide solution to give the corresponding alcohol and acid ?

- (1) Phenol
- (2) Benzoic acid
- (3) Butanal
- (4) Benzaldehyde

70. The compound formed on heating chlorobenzene with chloral in the presence of concentrated sulphuric acid is

- (1) gammexene
- (2) DDT
- (3) freon
- (4) hexachloroethane.

71. In the following sequence of reactions,



The compound 'D' is :-

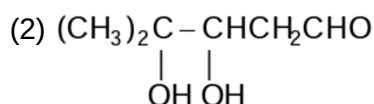
- (1) n-propyl alcohol
- (2) propanal
- (3) butanal
- (4) n-butyl alcohol

72. Iodoform can be prepared from all except :

- (1) Ethyl methyl ketone
- (2) Isopropyl alcohol
- (3) 3-Methyl-2-butanone
- (4) Isobutyl alcohol

73. On vigorous oxidation by permanganate solution $(\text{CH}_3)_2\text{C} = \text{CHCH}_2\text{CHO}$ gives

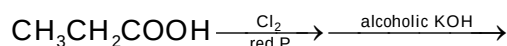
- (1) $(\text{CH}_3)_2\text{CO}$ and OHCCH_2CHO



- (3) $(\text{CH}_3)_2\text{CO}$ and $\text{OHCCH}_2\text{COOH}$

- (4) $(\text{CH}_3)_2\text{CO}$ and $\text{CH}_2(\text{COOH})_2$

74. End product of the following reaction is :

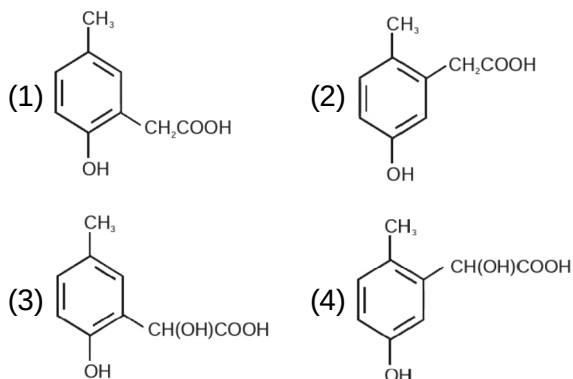


- (1) $\begin{array}{c} \text{CH}_3\text{CHCOOH} \\ | \\ \text{OH} \end{array}$
- (2) $\begin{array}{c} \text{CH}_2\text{CH}_2\text{COOH} \\ | \\ \text{OH} \end{array}$

- (3) $\text{CH}_2=\text{CHCOOH}$
- (4) $\begin{array}{c} \text{CH}_2\text{CHCOOH} \\ | \quad | \\ \text{Cl} \quad \text{OH} \end{array}$

SPACE FOR ROUGH WORK

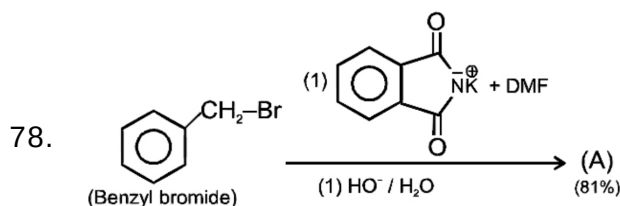
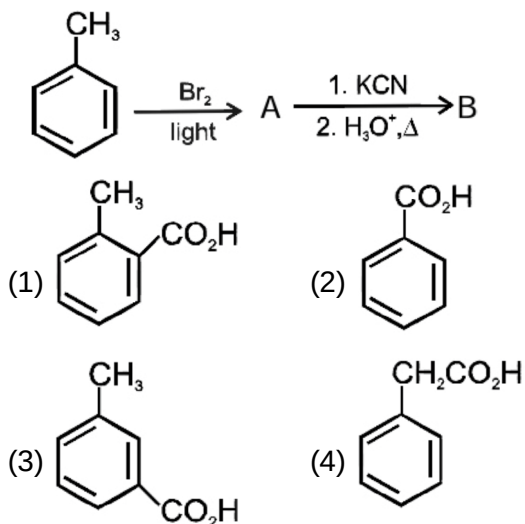
75. p-cresol reacts with chloroform in alkaline medium to give the compound A which adds hydrogen cyanide to form, the compound B. The latter on acidic hydrolysis gives chiral carboxylic acid. The structure of the carboxylic acid is :



76. A liquid was mixed with ethanol and a drop of concentrated H_2SO_4 was added. A compound with a fruity smell was formed. The liquid was :



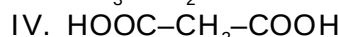
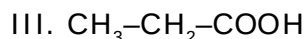
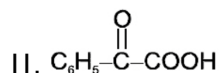
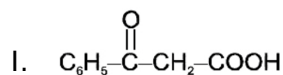
77. What is the final product (B) of this reaction sequence ?



Product (A) of the above reaction is :

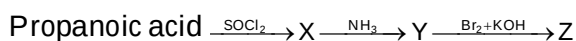
- (1) Ph-NH_2
(2) $\text{Ph-CH}_2\text{-NH}_2$
(3) $\text{Ph-CH}_2\text{-NH-CO}_2\text{H}$
(4) $\text{Ph-CH}_2\text{-NH-CHO}$

79. For the following acids the rate of decarboxylation on heating would be :



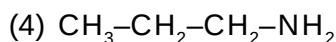
- (1) $\text{III} > \text{I} > \text{IV} > \text{II}$ (2) $\text{I} > \text{III} > \text{IV} > \text{II}$
(3) $\text{III} > \text{IV} > \text{I} > \text{II}$ (4) $\text{I} > \text{IV} > \text{II} > \text{III}$

80. Starting from propanoic acid, the following reactions were carried out



What is the compound Z

- (1) $\text{CH}_3\text{-CH}_2\text{-Br}$
(2) $\text{CH}_3\text{-CH}_2\text{-NH}_2$
(3)

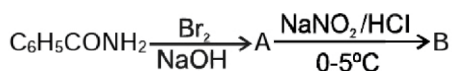


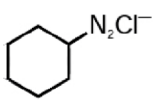
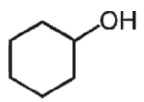
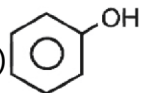
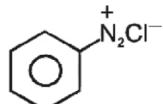
81. Which of the following acids remains unaffected on heating

- (1) Malonic acid (2) Malic acid
(3) Fumaric acid (4) Succinic acid

SPACE FOR ROUGH WORK

82. The product B of the following sequence of reactions is



- (1)  (2) 
(3)  (4) 

83. Hoffmann bromamide degradation reaction is shown by _____.

- (1) ArNH_2 (2) ArCONH_2
(3) ArNO_2 (4) ArCH_2NH_2

84. In order to prepare a 1° amine from an alkyl halide with simultaneous addition of one CH_2 group in the carbon chain, the reagent used as source of nitrogen is _____.

- (1) Sodium amide, NaNH_2
(2) Sodium azide, NaN_3
(3) Potassium cyanide, KCN
(4) Potassium phthalimide, $\text{C}_6\text{H}_4(\text{CO})_2\text{N}^-\text{K}^+$

85. Best method for preparing primary amines from alkyl halides without changing the number of carbon atoms in the chain is

- (1) Hoffmann Bromamide reaction
(2) Gabriel phthalimide synthesis
(3) Sandmeyer reaction
(4) Reaction with NH_3

86. The product formed by the reaction of acetamide with Br_2 in presence of NaOH is :

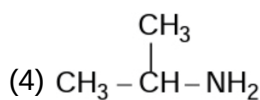
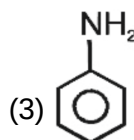
- (1) CH_3CN (2) CH_3CHO
(3) $\text{CH}_3\text{CH}_2\text{OH}$ (4) CH_3NH_2

87. $\text{C}_6\text{H}_5\text{NH}_2 \xrightarrow[0-5^\circ\text{C}]{\text{NaNO}_2+\text{HCl}} \text{X} \xrightarrow[\Delta]{\text{H}_2\text{O}} \text{Y}$,
the product Y is :

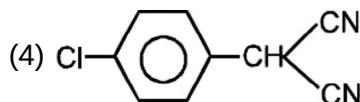
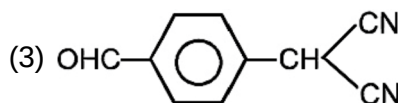
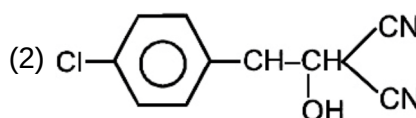
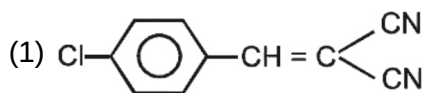
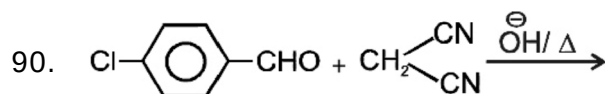
- (1) Benzenediazonium chloride
(2) Nitrobenzene
(3) Phenol
(4) Cresol

88. Which of the following will not give positive test with CHCl_3/KOH .

- (1) $\text{CH}_3\text{CH}_2\text{NHCH}_3$
(2) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$



89. The Hinsberg's method is used for:
(1) preparation of primary amines
(2) preparation of secondary amines
(3) preparation of tertiary amines
(4) separation of amine mixtures



SPACE FOR ROUGH WORK

91. Restriction endonucleases
- (1) cleave DNA at highly specific recognition sequences
 - (2) are inserted into bacteria by bacteriophages
 - (3) contain axon and intron
 - (4) add methyl groups to specific DNA sequences
92. X technique is now routinely used to detect HIV in suspected AIDS patients. It is being used to detect mutations in genes in suspected cancer patients too. It is a powerful technique to identify many other genetic disorders Identify X
- (1) X = PCR
 - (2) X = DNA fingerprinting
 - (3) X = Pathogen
 - (4) X = X-ray defraction
93. For transformation with recombinant DNA, the bacterial cells must first be made 'competent' which means
- (1) should increase their catalytic activity
 - (2) should decrease their metabolic reactions
 - (3) increases efficiency with which DNA enters the bacterium
 - (4) ability to divide fast
94. During isolation of DNA, addition of which of the following causes precipitation of purified DNA ?
- (1) Chilled ethanol
 - (2) RNA primer
 - (3) DNA polymerase
 - (4) Proteases
95. A bioreactor (fermenter) refers to
- (1) a device in which substances are treated to stimulate biochemical transformation by living cells
 - (2) study of mutated species
 - (3) a tank for biochemical reactions
 - (4) organisms reacting to a stimulus

96.

Column-I		Column-II	
I	PCR	A	Large scale culture
II	Bioreactor	B	To induce alien DNA in host cell
III	Genegun	C	Restriction endonuclease
IV	EcoR1	D	Amplification of gene

- (1) I-D, II-A, III-B, IV-C
- (2) I-B, II-A, III-D, IV-C
- (3) I-C, II-B, III-D, IV-A
- (4) I-A, II-D, III-B, IV-C

97.

Column-I		Column-II	
I	Band	A	Restriction fragments are attached to this
II	Positive pole	B	Where specific restriction fragment collects in gel
III	Taq polymerase	C	Piece of DNA cut up by restriction enzymes
IV	Recognition	D	Temperature resistant

- (1) I-D, II-A, III-B, IV-C
- (2) I-A, II-D, III-B, IV-C
- (3) I-D, II-A, III-C, IV-B
- (4) I-B, II-A, III-D, IV-C

98.

Column-I		Column-II	
I	EcoRI	A	<i>Bacillus amyloliquefaciens</i>
II	Bam HI	B	<i>Haemophilus influenza</i>
III	Hind III	C	<i>Escherichia coli</i>
IV	pBR322	D	Artificial plasmid

- (1) I-C, II-A, III-B, IV-D
- (2) I-B, II-A, III-D, IV-C
- (3) I-D, II-A, III-C, IV-B
- (4) I-A, II-D, III-B, IV-C

99. Which one of these statements about the applications of gene cloning is false ?

- (1) Large amounts of recombinant protein can be produced by gene cloning
- (2) DNA fingerprinting is used to detect proteins bound to DNA
- (3) Cloned genes can be used to detect carrier of disease-causing genes
- (4) Gene therapy attempts to correct a disorder by delivering a good copy of a gene to a patient

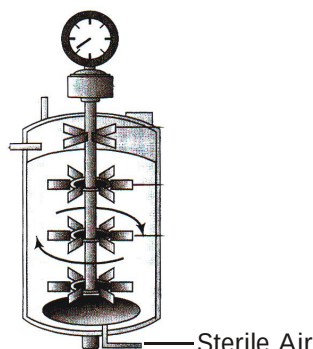
100. In order to identify the person who committed a crime, forensic experts will need to extract DNA from the tissue sample collected at the crime scene and conduct one of the following procedures for DNA finger printing analysis.

- (1) Cut the DNA and hybridize with specific micro-satellite probes
- (2) Cut the DNA and subclone the fragments
- (3) Cut DNA of victim to transfer in other individual
- (4) (2) followed by (3)

101. Select the correct option which shows the most appropriate temperatures of three different steps of PCR mechanism

Denaturation	Primer annealing	Primer extension
(1) 40–60°C	72°C	90°C
(2) 60–60°C	96°C	40–60°C
(3) 94–96°C	40–60°C	72°C
(4) 27°C	45–50°C	85–87°C

102. Identify the correct match for the given apparatus.



Apparatus		Function	
(1)	Genegun	i	Vectorless direct gene transfer
(2)	Paper chromatography	ii	Separation of chlorophyll pigments
(3)	Stirred tank bioreactor	iii	Carry out fermentation process
(4)	Respirometer	iv	Finding out rate of respiration

103. Which of the following statement is not true ?

- (1) Eco RI always cuts DNA molecules at a particular point by recognizing a specific sequence of 4 base pairs
- (2) Besides Hind-II, today we know more than 900 restriction enzymes
- (3) restriction enzymes are part of defence mechanism
- (4) Type-II restriction endonuclease is most useful in genetic engineering

104. Antibiotic resistance gene present of Bam HI site of a *E.coli* cloning vector is

- (1) streptomycin resistance
- (2) tetracycline resistance
- (3) chloramphenicol resistance
- (4) kanamycin resistance

105. Direct visual selection method for the selection of recombinant host cells on the basis of their inability to produce colour in the presence of chromogenic substrate can be applied, if the vector used is a

- (1) modified plasmid vector of *E.coli*
- (2) mutated phages
- (3) modified *Agrobacterium tumefaciens* plasmid
- (4) disarmed retrovirus

106. Transgenic animal has

- (1) foreign DNA in all its cells
- (2) foreign RNA in all its cells
- (3) foreign DNA in some of the cells
- (4) all of the above

107. There are three basic steps in genetically modify organism. Arrange these steps in correct sequence.
- Introduction of the identified DNA into the host.
 - Maintenance of introduced DNA in the host and transfer of the DNA to its progeny.
 - Identification of DNA with desirable genes.
- $a \rightarrow b \rightarrow c$
 - $b \rightarrow c \rightarrow a$
 - $c \rightarrow b \rightarrow a$
 - $c \rightarrow a \rightarrow b$
108. In the year ___a___ the two enzymes responsible for ___b___ the growth of bacteriophage in *E.coli* were isolated. One of these added ___c___ group to DNA, while other ___d___ DNA. The later was called ___e___.
- a-1967, b-promoting, c-ethyl, d-join, e-DNA ligase
 - a-1963, b-restricting, c-methyl, d-join, e-DNA ligase
 - a-1972, b-restricting, c-methyl, d-cut, e-restriction endonuclease
 - a-1963, b-restricting, c-methyl, d-cut, e-restriction endonuclease
109. The most important feature in a plasmid to be used as a vector is
- Origin of replication
 - Presence of sites for restriction endonuclease
 - Presence of selectable marker
 - Presence of alternate selectable marker
110. In vector pBR 322, the 'rop' codes for the
- Pvu II
 - Pvu I
 - Proteins that involved in the replication of the restriction enzyme
 - Proteins that involved in the replication of the plasmid
111. Which statement is incorrect?
- Recognition sequence is made up of 6 bases
 - Recognition sites are present in cloning vector
 - In gene-gun, plant cell are bombarded with high velocity micro-particles of gold or tungsten
 - Agrobacterium tumifaciens* is a pathogen of several dicot plants
112. Restriction enzymes belongs to a large class of the enzyme called
- Cellulase
 - Nuclease
 - Chitinase
 - Polymerase
113. In pBR 322, how many recognition sites are present?
- 8
 - 7
 - 3
 - 4
114. Stickiness of the sticky ends of the DNA facilitates the
- Action of DNA ligase and these ends joined together laterally
 - Action of DNA ligase and these ends joined together end-to-end
 - Action of Taq polymerase
 - Action of restriction enzyme
115. In pBR 322, recognition sequences, that are present on the same type of selectable marker.
- Pst I, Pvu I
 - Bam HI, Sal I
 - Both
 - None of the above
116. DNA is a ___1___ molecule having ___2___ charge.
- 1-Hydrophobic, 2-negative
 - 1-Hydrophilic, 2-positive
 - 1-Hydrophilic, 2-negative
 - 1-Hydrophobic, 2-positive
117. Source of Taq polymerase used in PCR is a
- Thermophilic fungus
 - Mesophilic fungus
 - Thermophilic bacterium
 - Halophilic bacterium

118. A typical bioreactor has
- An agitator system
 - An oxygen delivery system
 - A foam control system
 - A temperature control system
 - A pH control system
 - Sampling ports
- a, b and c
 - a, b, c and d
 - a, b, c, d and e
 - a, b, c, d, e and f

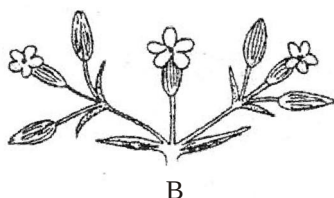
119. Significance of the 'heat shock' in bacterial transformation is to facilitate.
- Expression of the antibiotic resistance gene in the vector
 - Ligation of DNA to the cell membrane
 - Uptake of DNA through membrane transport protein
 - Uptake of DNA through pores in the bacterial cell wall

120. Select the incorrect statement(s)
- A foreign DNA can be ligated at the BamHI site of ampicillin resistance gene in the vector pBR 322.
 - Some plasmids may have only one or 2 copies per cell whereas others may have 15-100 copies per cell.
 - In almost all recombinant technologies, the ultimate aim is to produce a desirable protein.
 - All of the above

121. Identify the types of inflorescence shown in figure and select the correct option for A and B.



A



B

- | | |
|--------------|----------|
| A | B |
| (1) Cymose | Racemose |
| (2) Racemose | Cymose |
| (3) Racemose | Racemose |
| (4) Cymose | Cymose |

122. Roots are modified to perform specific functions other than their normal functions. The given figure shows modification of the roots of mangrove plant. Select the incorrect option regarding it.



- The stilt roots of red mangrove help in breathing.
 - The root system is highly entangled, huge and extensive under the water.
 - A large number of animals such as small fishes, crustaceans, seahorses, etc. find shelter in this root system.
 - Besides providing mechanical support, these roots also perform photosynthetic functions in the plant.
123. Consider the following statements.
- In *Gynandropis*, *Passiflora*, etc., thalamus is elongated and shows well developed nodes and internodes.
 - The buds in *Agave* sometimes get modified into bulbils.
 - Sepals are concerned with protection of flowers in bud condition and petals help to attract insects for pollination.
 - Stamens and carpels serve as the male and female reproductive organs respectively.

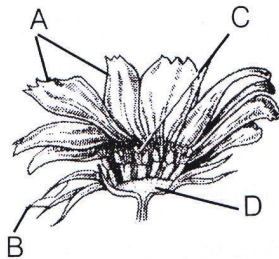
Which of the following combinations of above statements provides an evidence that flower is a modified shoot?

- (i) and (ii)
 - (ii) and (iii)
 - (iii) and (iv)
 - (i) and (iv)
124. Pneumatophodes are morphologically equivalent to
- stomata
 - lenticels
 - hydathodes
 - wounds

125. Which one of the following sets represents the example of stilt roots?

- (1) *Bryophyllum*, *Begonia*, *Loranthus*
- (2) *Dahlia*, *Asparagus*, *Pandanus*
- (3) *Cuscuta*, *Loranthus*, *Bryophyllum*
- (4) *Zea mays*, *Saccharum*, *Pandanus*

126. The following figure shows the LS of inflorescence of *Helianthus annuus* flower. Find out the correct sequence of labelling.



- (1) A-Disc floret; B-Receptacle; C-Ray florets; D -Involucre
- (2) A-Ray florets; B-Involucre; C-Disc florets; D-Receptacle
- (3) A-Disc florets; B-Involucre; C-Receptacle; D-Ray florets
- (4) A-Ray florets; B-Disc florets; C-Receptacle; D-Involucre

127. Identify the correct matched pair.

- (1) Turmeric - Tuber
- (2) Potato - Rhizome
- (3) Onion - Tunicated bulb
- (4) *Helianthus tuberosus* - bulb

128. Identify the correctly matched option -

- (1) Modification of stem - Corn of *Colocasia*
- (2) Cladode - *Opuntia*
- (3) Phylloclade - *Euphorbia*
- (4) Lianas - Arctic region

129. Which of the following is a wrong statement?

- (1) Rose has prickles
- (2) In *Ficus*, parallel venation is found
- (3) *Solanum* is polytypic genus
- (4) *Tulips* are the members of lily family

130. Which of the following is not correct?

- (1) Carrot and turnip - modified tap root
- (2) Sweet potato - Prop root
- (3) Maize and sugarcane - Stilt root
- (4) *Rhizophora* - Pneumatophores

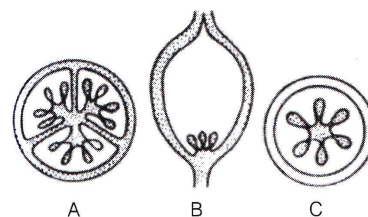
131. Which of the following is not correct?

- (1) When veinlets form a network, the venation is termed as reticulate
- (2) When the veins run parallel to each other within lamina the venation is termed as parallel
- (3) Parallel venation is a characteristic of dicots
- (4) Reticulate venation is a characteristic of dicots

132. Which one of the following matches is correct with respect to placentation?

- (1) Pea - Axile
- (2) *Dianthus* - Free central
- (3) *Sunflower* - Parietal
- (4) Arhar - Basal

133. Identify the placentation from the given diagrams and give one example of each -



- (1) A-Marginal, Pea; B-Parietal, Argemone; C-Basal, Wheat
- (2) A-Axile, Citrus; B-Basal, Wheat; C-Free central, Primrose
- (3) A-Axile, Citrus; B-Basal, Wheat; C-Free central, Rose
- (4) A-Parietal, Radish; B-Basal; Maize; C-Superficial; Nymphaea

134. Glumes are

- (1) calyx like, green leafy bracteoles present in whorls or whorls below the calyx occurring in most members
- (2) small dry scaly bracts found in spikelet of the members of family - Poaceae
- (3) green - coloured and in one or more whorls around and below the entire inflorescence
- (4) brightly coloured appearing like the petals of flower

135. Adventitious roots are adventitious in their :-

- (1) function
- (2) position
- (3) place of origin
- (4) internal structure

136. A few millimeters above the root cap is the region of :-

- (1) elongation
- (2) maturation
- (3) meristematic activity
- (4) root hairs

137. One of the following is not a root vegetable?

- (1) Sweet potato
- (2) Potato
- (3) Turnip
- (4) Carrot

138. Thorn develops from :-

- (1) Root
- (2) Leaf
- (3) Axillary bud
- (4) peduncle

139. Match column I with column II.

	Column-I		Column-II
A	Bud in the axil of	I	Pitcher plant and Venus Fly trap
B	Bud in the axil of	II	Cacti
C	Spines (modified leaves)	III	Compound leaf
D	Leaves modified to catch insects	IV	Simple leaf
E	Fleshy leaves with stored food	V	Garlic and onion

- (1) A-I; B-II; C-III; D-IV; E-V
- (2) A-V; B-IV; C-III; D-II; E-I
- (3) A-IV; B-III; C-II; D-I; E-V
- (4) A-IV; B-II; C-III; D-I; E-V

140. Racemose inflorescence is identified by :-

- (1) acropetal arrangement of flowers on peduncle
- (2) presence of sessile flowers
- (3) continuous growth of main axis
- (4) Both 1 and 3

141. Cymose inflorescence is identified by :-

- (1) Basipetal arrangement of flowers on the main axis (peduncle)
- (2) The limited growth of the main axis as main axis terminates in a flower
- (3) Both 1 and 2
- (4) Presence of sessile flower

142. Match the Column-I with Column-II.

	Column-I		Column-II
A	Gamosepalous	I	Flower of lily
B	Polysepalous	II	Sterile anther
C	Gamopetalous	III	Free petals
D	Polypetalous	IV	Free sepals
E	Epiphyllous	V	Fused petals
F	Staminode	VI	Fused sepals

- (1) A-IV; B-V; C-III; D-I; E-VI; F-II
- (2) A-IV; B-V; C-III; D-I; E-II; F-VI
- (3) A-VI; B-IV; C-V; D-III; E-I; F-II
- (4) A-VI; B-IV; C-V; D-III; E-II; F-I

143. Which of the following combinations is false?

- (1) Apocarpous - Carpels free - Lotus, Rose
- (2) Syncarpous - Carpels fused - Mustard, tomato
- (3) Placentation - arrangement of ovules within ovary
- (4) Arrangement of ovules within ovary - ovulation

144. Which of the following is not non-endospermous seed (in matured condition) ?

- (1) Bean, gram, pea
- (2) Coconut, cereals (e.g. maize)
- (3) Both 1 and 2
- (4) Mango

145. Scutellum is the first leaf of

- (1) monocot
- (2) dicot
- (3) gymnosperm
- (4) pteridophytes

146. Match the column-I and column-II with column-III.

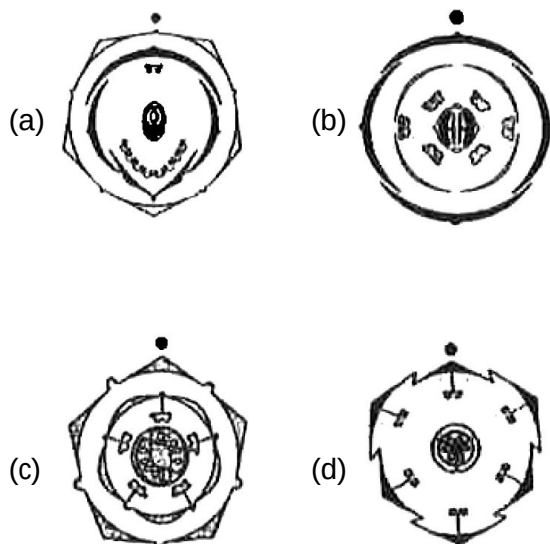
Column-I

- (A) Brassicaceae (B) Fabaceae
(C) Solanaceae (D) Liliaceae

Column-II

- (i) $\text{Br} \oplus \text{P}_{3+3} \text{A}_{3+3} \text{G}_{(3)}$
(ii) $\text{Br} \oplus \text{K}_{(5)} \text{C}_{(5)} \text{A}_5 \text{G}_3$
(iii) $\% \text{K}_{(5)} \text{C}_{1+2+(2)} \text{A}_{(9)+1} \text{G}_1$
(iv) $\oplus \text{K}_{2+2} \text{C}_4 \text{A}_{2+4} \text{G}_{(2)}$

Column-III



- (1) A-iv-b; B-iii-a; C-ii-c; D-i-d
(2) A-iv-a; B-iii-b; C-ii-c; D-i-d
(3) A-iv-b; B-iii-a; C-ii-c; D-i-d
(4) A-iv-b; B-iii-c; C-ii-a; D-i-d

147. Androeceum in *Hibiscus* is :-

- (1) Didynamous and monotheous
(2) Monodelphous and monotheous
(3) Diadelphous and bitheous
(4) Polyadelphous and monotheous

148. Which of the following is a flowering plant with nodules containing filamentous nitrogen fixing microorganism?

- (1) *Cicer* (2) *Gossypium*
(3) *Crotalaria* (4) *Cycas*

149. A method of cellular defence that takes place in all eukaryotic organisms is

- (1) RNAi
(2) downstream processing
(3) extension
(4) annealing

150. The process by which multiple copies of the gene of interest is synthesised in vitro using two of primers is known as

- (1) PCR
(2) gel electrophoresis
(3) downstream processing
(4) genetic printing

151. Now-a-days, the most commonly used matrix in gel electrophoresis is

- (1) agarose
(2) ethidium bromide
(3) sulphate bromide
(4) sulphite bromide

152. When an alien DNA is linked with the origin of replication, so that this alien piece of DNA can replicate and multiply itself in the host organism, this can be called as

- (1) gene transfer
(2) gel electrophoresis
(3) cloning
(4) transformation

153. In recombinant DNA technology, the plasmid vector DNA is cleaved by

- (1) ribonuclease (2) exonuclease
(3) endonuclease (4) any nuclease

154. The bacterial resistance to antibiotics is a genetic qualification present in

- (1) virus
(2) bacterial DNA
(3) bacterial membrane
(4) plasmid

155. Plasmid that has the ability of integrating into the bacterial DNA are called

- (1) nucleotides (2) hyposomes
(3) virion (4) episome

156. A gene carried by recombinant DNA is cloned when
- (1) its host bacterium divides by binary fission
 - (2) it is transcribed
 - (3) it is fragmented by restriction enzymes
 - (4) it is hybridised
157. DNA of plasmid is
- (1) double-stranded and circular
 - (2) single-stranded and circular
 - (3) double-stranded and linear
 - (4) single-stranded and linear
158. In certain retroviruses, have the ability to transform normal animal cell into
- (1) dead cell
 - (2) RNA containing cell
 - (3) DNA containing cell
 - (4) cancerous cell
159. The Father of DNA fingerprinting is
- (1) Sunder Lai Bahuguna
 - (2) Vishwanath
 - (3) Jeffreys
 - (4) Rockfeller
160. Genetically identical copy of individual cell or gene is known as
- (1) species
 - (2) phagemid
 - (3) clone
 - (4) plasmid
161. The technique of breakage of DNA fragment and inserting it into another DNA molecule, is related to
- (1) gene splicing
 - (2) gene cloning
 - (3) gene typing
 - (4) DNA fingerprinting
162. The number of copies of a gene present in a cell is called
- (1) gene dosage
 - (2) gene pool
 - (3) gene amplification
 - (4) gene frequency
163. Type-I restriction enzyme
- (1) recognise the specific sequence and cleave DNA at the specific sequence
 - (2) do not recognise or cleave the specific sequence
 - (3) recognise specific sequence but cleave non-specific sequence
 - (4) None of the above
164. Which of the following is not associated with genetic engineering?
- (1) Phage
 - (2) Plasmids
 - (3) Mycorhiza
 - (4) Transgenesis
165. VNTRs are
- (1) Variable Number Tandem Repeats
 - (2) Very Narrow Tandem Repeats
 - (3) Variable Non-cistronic Transposon Repeats
 - (4) Valuable Non-cistronic Transposic Regions
166. In a chromosome, there is a specific DNA sequence, which is responsible for initiating replication, known as
- (1) vector
 - (2) origin of replication
 - (3) recognition sequence
 - (4) marker gene
167. The cutting of DNA by restriction endonucleases results in the fragments of DNA. These fragments can be separated by a technique known as
- (1) PCR
 - (2) SDS page
 - (3) Northern blotting
 - (4) Gel electrophoresis
168. You can see bright orange coloured bands of DNA in a ethidium bromide stained gel exposed to
- (1) X-rays
 - (2) sun rays
 - (3) UV rays
 - (4) gamma rays

169. A thermostable DNA polymerase, which remains active during the high temperature induced denaturation of double-stranded DNA is isolated from a bacterium

- (1) *E. coli*
- (2) *Thermos aquaticus*
- (3) *Salmonella paratyphi*
- (4) *Thermus aquaticus*

170. Which one of the following is a false fruits?

- (1) Brinjal
- (2) Coconut
- (3) Apple
- (4) Tomato

171. Pepo is a fruit of

- (1) Cruciferae
- (2) Leguminosae
- (3) Cucurbitaceae
- (4) Liliaceae

172. Marginal placentation is found in

- (1) *Pisum* and *Lathyrus*
- (2) *Citrus* and *Solanum*
- (3) *Triticum* and *Ranunculus*
- (4) *Zea mays* and *Nymphaea*

173. Apocarpous gynoecium is found in

- (1) *Ranunculaceae*
- (2) *Malvaceae*
- (3) *Solanaceae*
- (4) *Papaveraceae*

174. Perianth is

- (1) outer part of flower in which calyx and corolla are not always distinguishable
- (2) when stamens and gynoecium are not distinguishable
- (3) when only calyx is present
- (4) None of the above

175. From of the following contains all others structures given in gynoecium except

- (1) ovary
- (2) stigma
- (3) carpel
- (4) filament

176. A pome fruit is said to be false because

- (1) the pericarp is in conspicuous
- (2) the endocarp is cartilaginous
- (3) the fruit is present in fleshy of edible thalamus
- (4) the fruit is derived from inferior ovary

177. An inflorescence always forms a

- (1) multiple or composite fruit
- (2) simple fruit
- (3) dry dehiscent fruit
- (4) aggregate fruit

178. Match the following columns.

Column I		Column II	
A.	Marginal placentation	1	<i>Petunia</i>
B.	Axile placentation	2	<i>Dianthus</i>
C.	Free central placentation	3	<i>Mustard</i>
D.	Parietal placentation	4	<i>Pea</i>

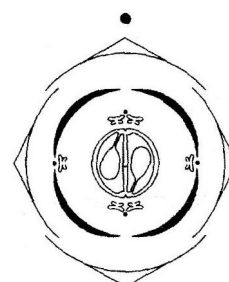
Codes :

A	B	C	D
(1) 2	1	4	3
(2) 3	4	2	1
(3) 4	1	2	3
(4) 4	1	3	2

179. Seed is a

- (1) fertilised ovary
- (2) fertilised fruit
- (3) fertilised embryo
- (4) fertilised ovule

180. Which of the following features characterize the family represented by the given floral diagram?



- (1) Cruciform corolla with quincuncial aestivation
- (2) Stamens with didynamous condition
- (3) Bicarpellary, syncarpous ovary with parietal placentation
- (4) Inflorescence usually cymose

Space For Rough Work
